INTRODUCTION TO INFORMATION SECURITY

WEEK3

CLO Covered in this Lecture:

CLO1: Explain key information security concepts and legal, ethical, and professional issues in information security.

What is Cryptography

- Data must be encrypted using a method that only parties with the proper authorization may decipher.
- It shouldn’t be simple for third parties to read it in case of any leakage or assault.
- In general, the cryptography process changes plaintext (readable data) into ciphertext (enciphered data).
- The cipher is a set of two algorithms used to encrypt and decode data afterward.
- Furthermore, special keys have been used together with algorithms.

Some Basic Terminology

Plaintext:
The original intelligible message
Cipher text:
The transformed message
Cipher:
An algorithm for transforming an intelligible message into one that is unintelligible by transposition and/or substitution methods
Key:
Some critical information used by the cipher, known only to the sender& receiver
Encipher (encode):
The process of converting plaintext to cipher text using a cipher and a key

Some Basic Terminology

Decipher (decode):
The process of converting cipher text back into plaintext using a cipher and a key

Cryptanalysis:
The study of principles and methods of transforming an unintelligible message back into an intelligible message without knowledge of the key. It is also called code-breaking

Cryptology:
Both cryptography and cryptanalysis

Cryptography

Cryptographic systems are generally classified along 3 independent dimensions:

1. **Type of operations used for transforming plain text to cipher text**

All the encryption algorithms are based on two general principles: **substitution**, in which each element in the plaintext is mapped into another element, and **transposition**, in which elements in the plaintext are rearranged.

Cryptography

2. The number of keys used

If the sender and receiver use the same key then it is said to be **symmetric** key (or) single key (or) conventional encryption.

If the sender and receiver use different keys then it is said to be **public** key encryption.

Cryptography

3. How the plain text is processed

A **block cipher** processes the input and block of elements at a time, producing an output block for each input block.

A **stream cipher** processes the input elements continuously, producing output elements one at a time, as it goes along.

Cryptanalysis

The process of attempting to discover the **plaintext (X), the secret key (K), or both** is known as cryptanalysis.
The strategy used by cryptanalysis depends on the nature of the encryption scheme and the information available to the cryptanalyst.

Cryptanalytic Attacks

There are various types of cryptanalytic attacks based on the amount of information known to the cryptanalyst.

Cipher text only

Known plaintext

Chosen plaintext

Chosen cipher text

Cipher text only – A copy of cipher text alone is known to the cryptanalyst.

Known plaintext – The cryptanalyst has a copy of the cipher text and the corresponding plaintext.

Cryptanalytic Attacks

Chosen plaintext – The cryptanalysts gain temporary access to the encryption machine. They cannot open it to find the key, however; they can encrypt a large number of suitably chosen plaintexts and try to use the resulting cipher texts to deduce the key

Chosen cipher text – The cryptanalyst obtains temporary access to the decryption machine, uses it to decrypt several strings of symbols, and tries to use the results to deduce the key.

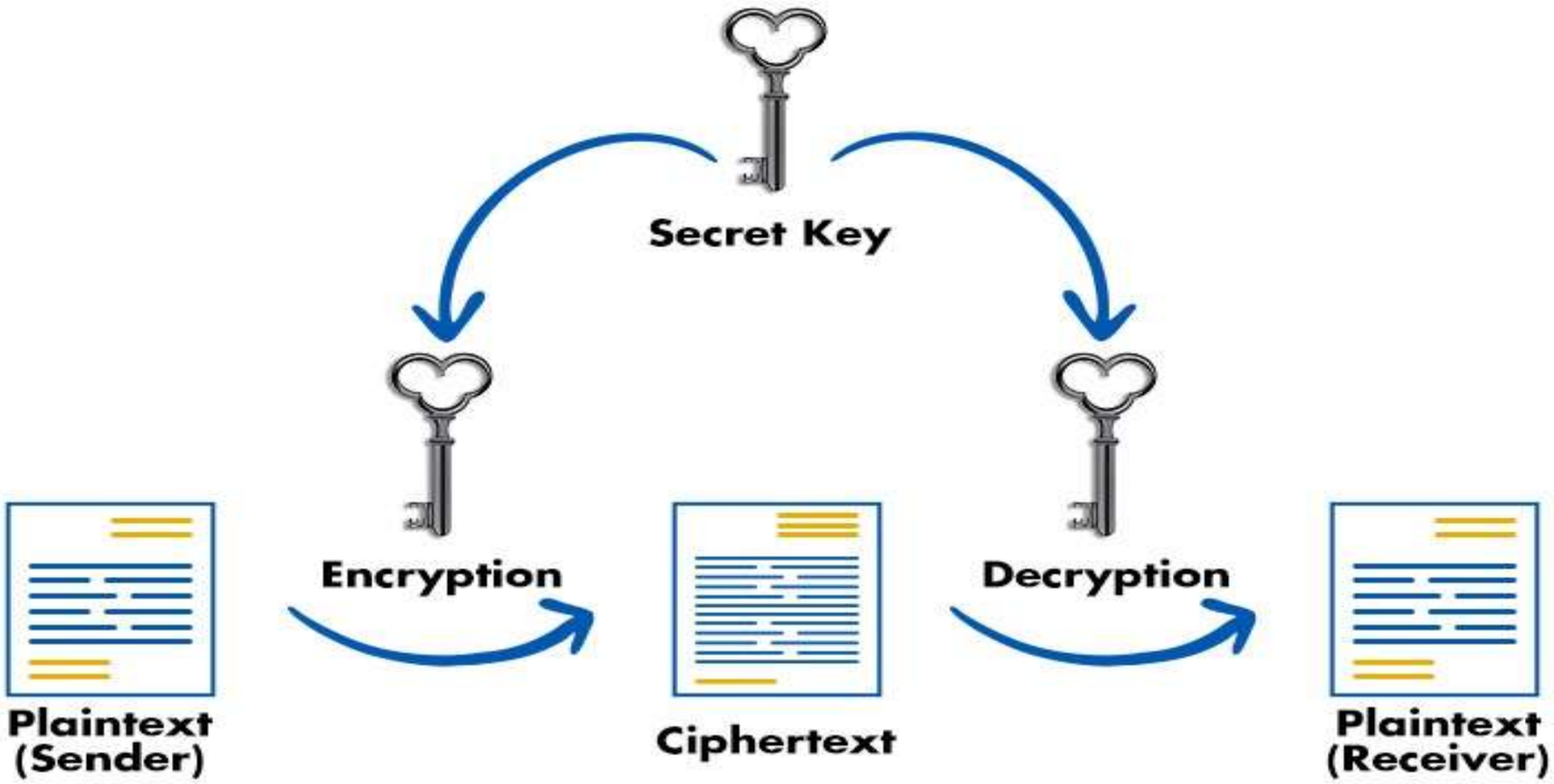
Types of Cryptography

- **Symmetric Cryptography**
- **Asymmetric Cryptography**

Symmetric Cryptography

- One key is used for both encryption and decryption in symmetric encryption.
- The algorithm and key combine to encrypt the original sensitive information by converting plaintext into ciphertext.
- This is effective for storing data that has to be decrypted at a later time.
- The use of a single key for encryption and decryption raises concerns because if the key got compromised, all the data it has encrypted would also be at risk.

Symmetric Cryptography



One such example of symmetric cryptography is bank transactions via payment applications. To prevent identity theft and fraudulent charges, personally identifiable information (PII) must be protected when used in payment applications.

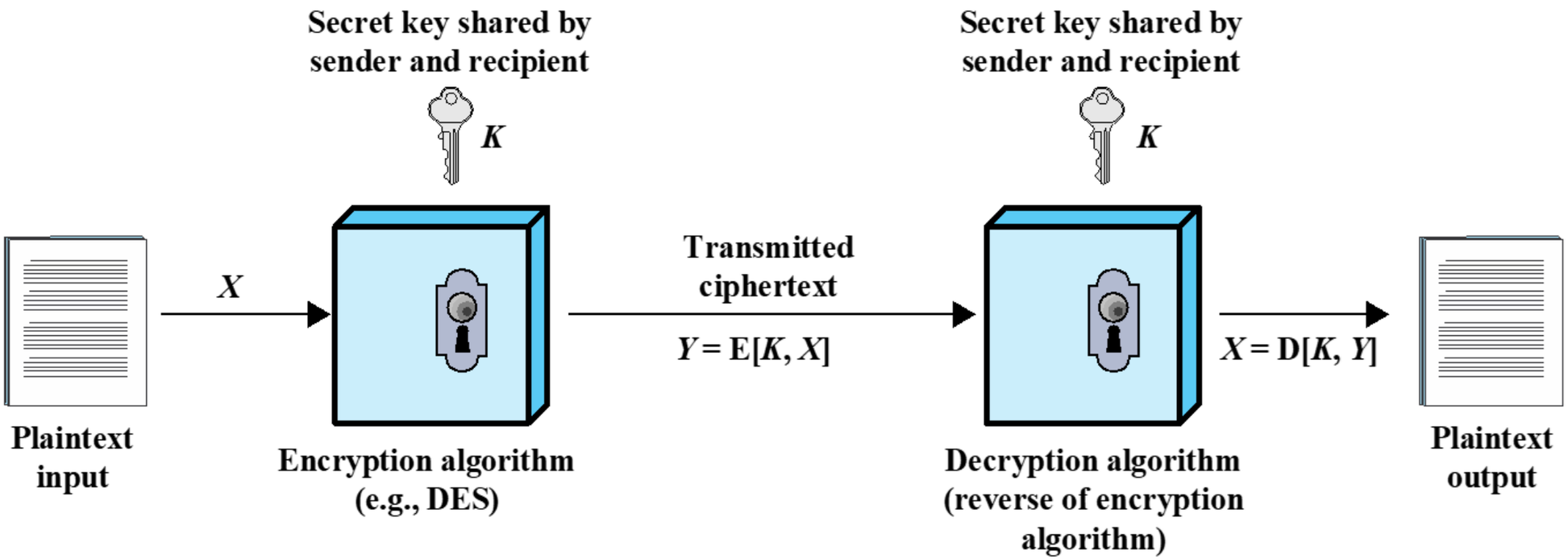


Figure 2.1 Simplified Model of Symmetric Encryption

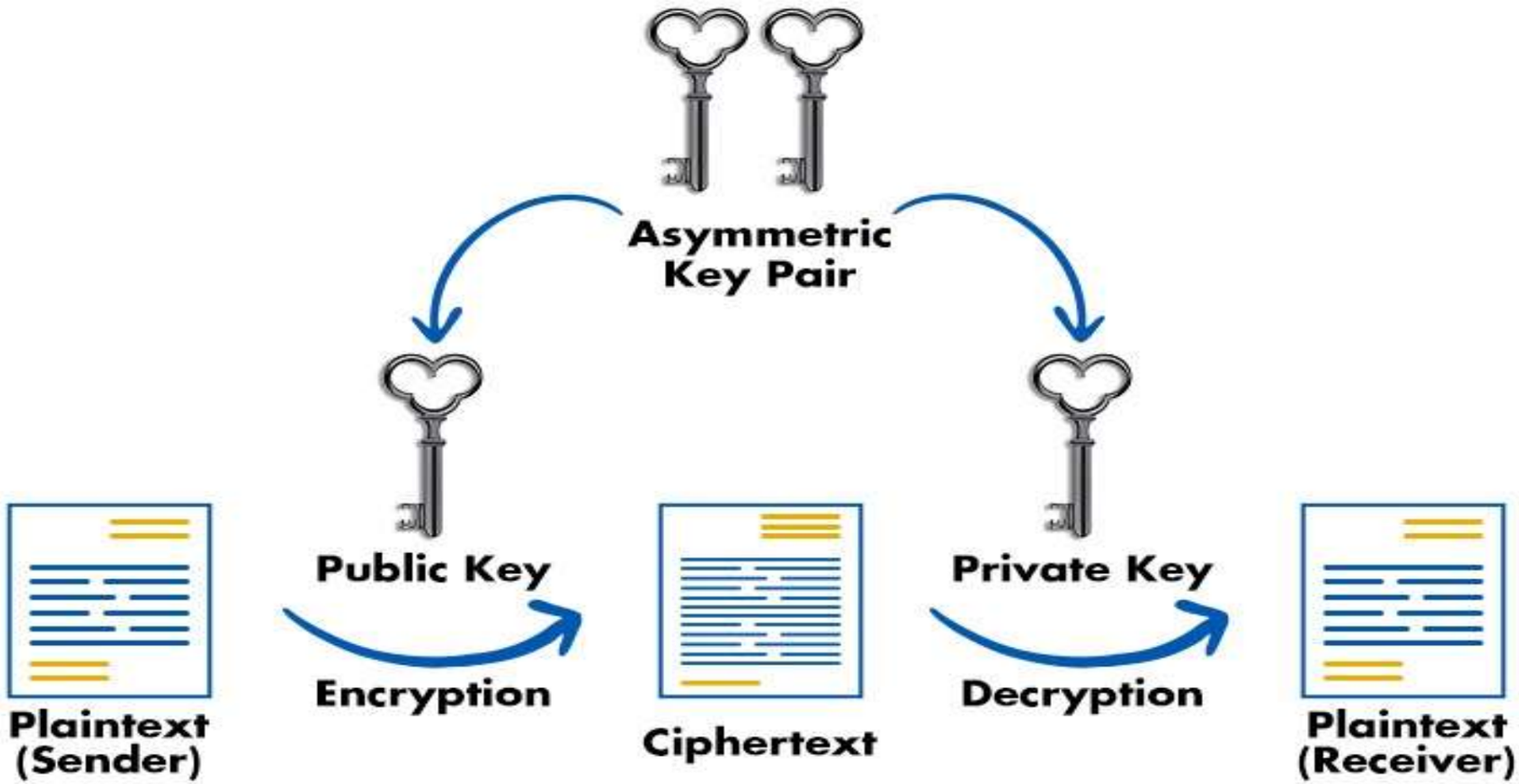
Asymmetric Cryptography

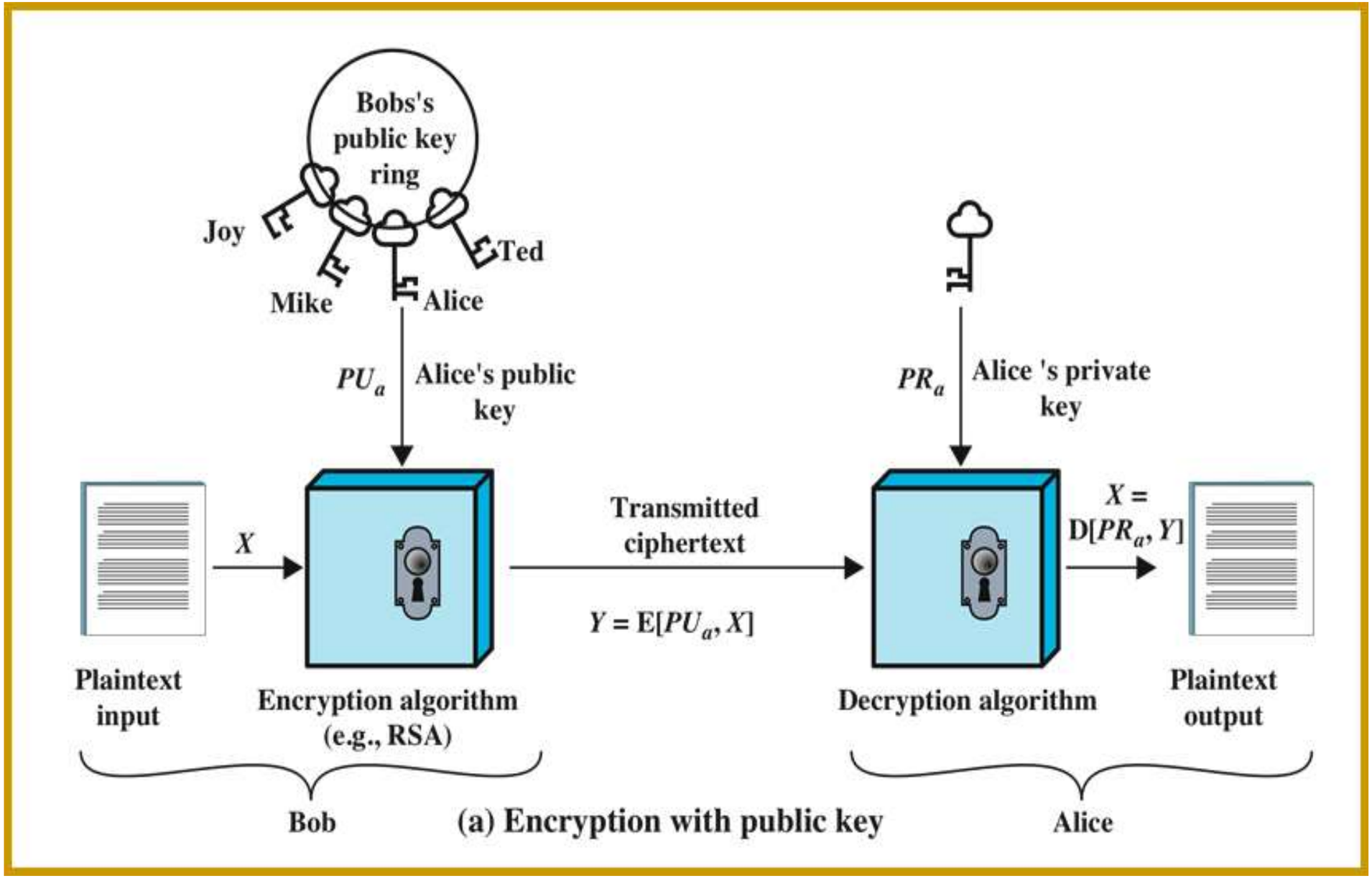
- Asymmetric cryptography, commonly called public key cryptography, is a more recent technology than symmetric encryption.
- A plain text is encrypted using two keys in asymmetric encryption.
- Over the Internet or a vast network, secret keys are exchanged. It prevents malicious individuals from misusing the keys.
- Asymmetric cryptography uses two related keys to increase security since it is crucial to keep in mind that anybody with access to the secret key may decrypt the message.
- Anybody who wants to send you a message may get a public key.
- The second private key is kept secret so that only you are aware of it.

Asymmetric Cryptography

- A message encrypted with a public key can only be decrypted with a private key.
- Since the public key is accessible to the whole public and may be sent via the Internet, security is not necessary.
- Asymmetric keys are far more effective in ensuring the confidentiality of data sent during transmission.

Asymmetric Cryptography





(a) Encryption with public key

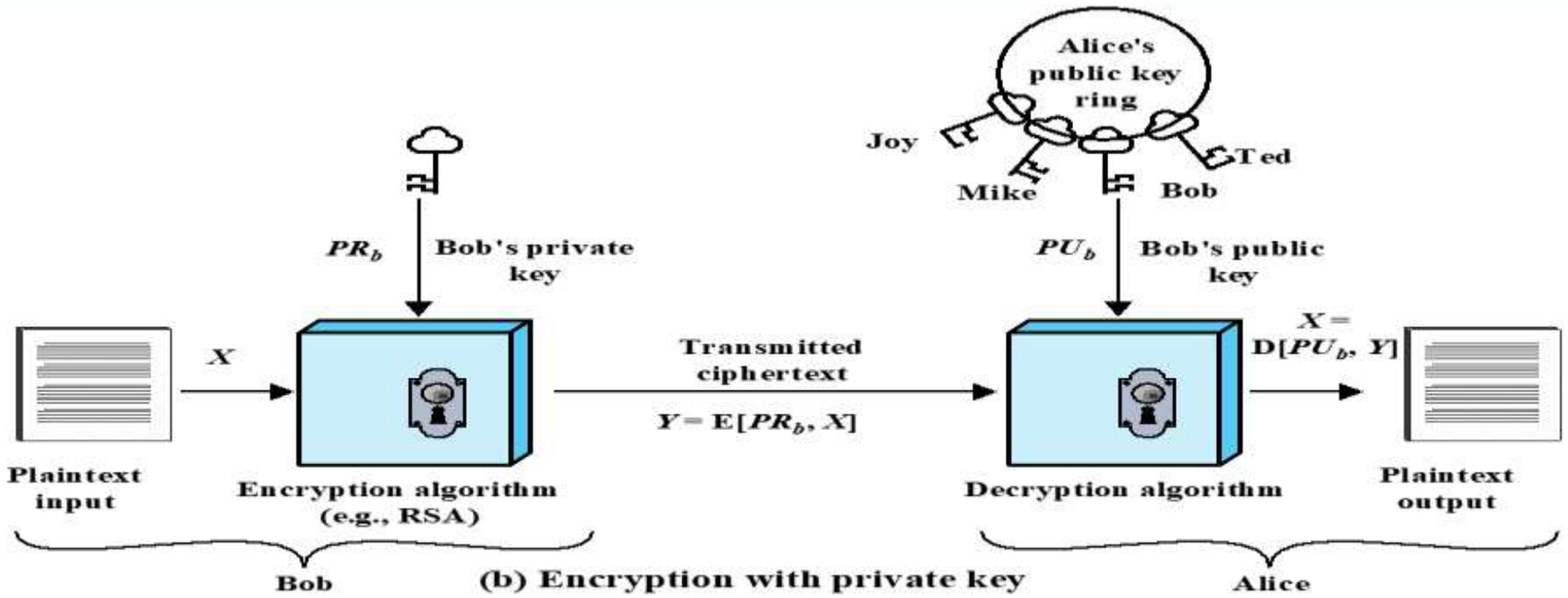


Figure 2.6 Public-Key Cryptography

- User encrypts data using his or her own private key
- Anyone who knows the corresponding public key will be able to decrypt the message

Requirements for Public-Key Cryptosystems

Computationally easy to create key pairs

Useful if either key can be used for each role

Computationally easy for sender knowing public key to encrypt messages

Computationally infeasible for opponent to otherwise recover original message

Computationally easy for receiver knowing private key to decrypt ciphertext

Computationally infeasible for opponent to determine private key from public key

- There are basically two types of symmetric cipher :
  - Substitution Cipher
  - Transposition Cipher

- **Substitution Cipher:** A substitution is a technique in which each letter or bit of the plaintext is substituted or replaced by some other letter, number or symbol to produce cipher text.

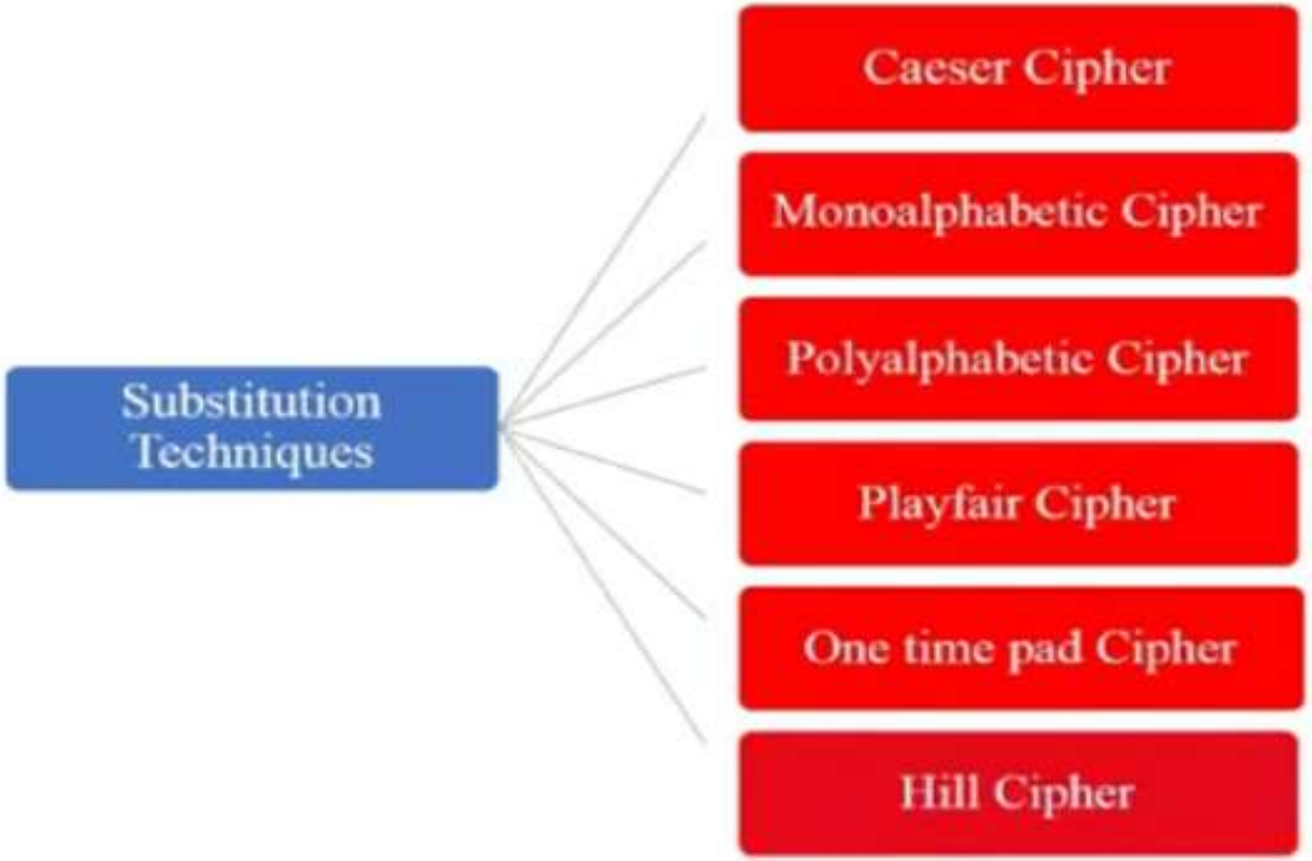
  For Example, ABC → XYZ

- **Transposition Cipher:** In transposition technique, there is no replacement of alphabets or numbers occurs instead their positions are changed or reordering of position of plain text is done to produce cipher text.

  For Example, ABCDE → BADEC

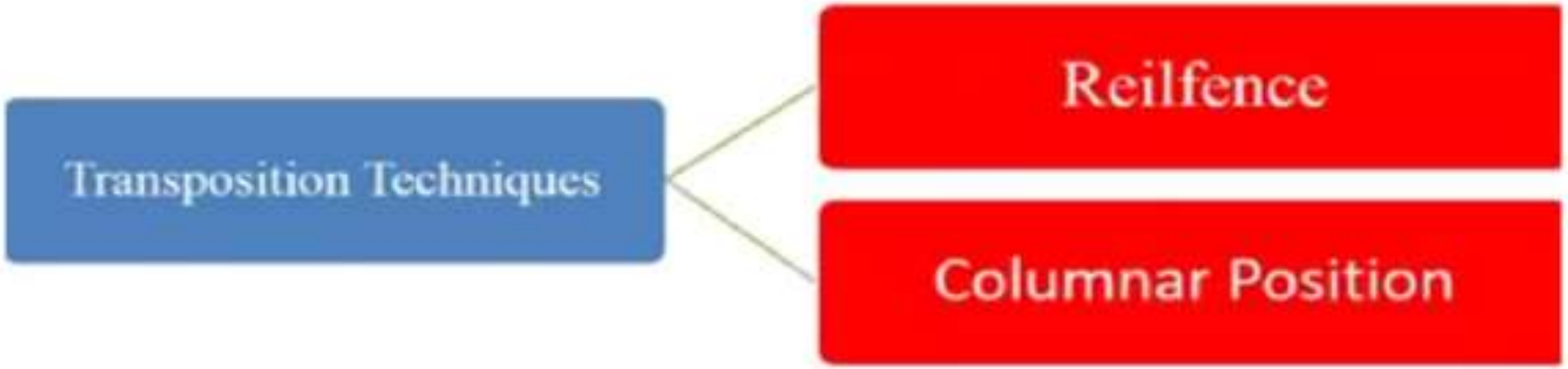
Substitution Cipher

- Types of substitution techniques:



Transposition Cipher

- Types of transposition techniques:



The Playfair Cipher Encryption Algorithm

Generate the key Square(5×5)

The key square is a 5×5 grid of alphabets that acts as the key for encrypting the plaintext. Each of the 25 alphabets must be unique and one letter of the alphabet (usually J) is omitted from the table (as the table can hold only 25 alphabets). If the plaintext contains J, then it is replaced by I.

- The initial alphabets in the key square are the unique alphabets of the key in the order in which they appear followed by the remaining letters of the alphabet in order.

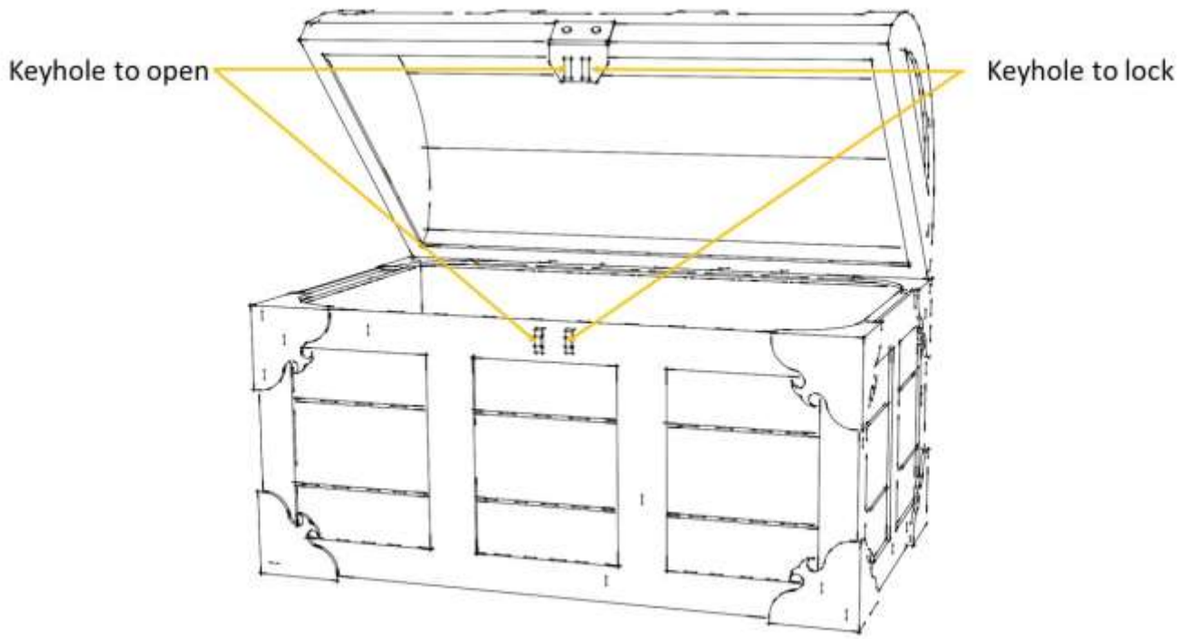
- **Algorithm to encrypt the plain text:** The plaintext is split into pairs of two letters (digraphs). If there is an odd number of letters, a Z is added to the last letter.
  - PlainText: "**instruments**"
  - After Split: **'in' 'st' 'ru' 'me' 'nt' 'sz'**
- Pair cannot be made with same letter. Break the letter in single and add a bogus letter to the previous letter.
  - Let's split "HELLO" into pairs: **HE LL O**
  - The second pair is "LL" (both letters are the same).To fix this, we break "LL" and insert a bogus letter (e.g., 'X' or 'Z').The new split will be:

HE LX LO

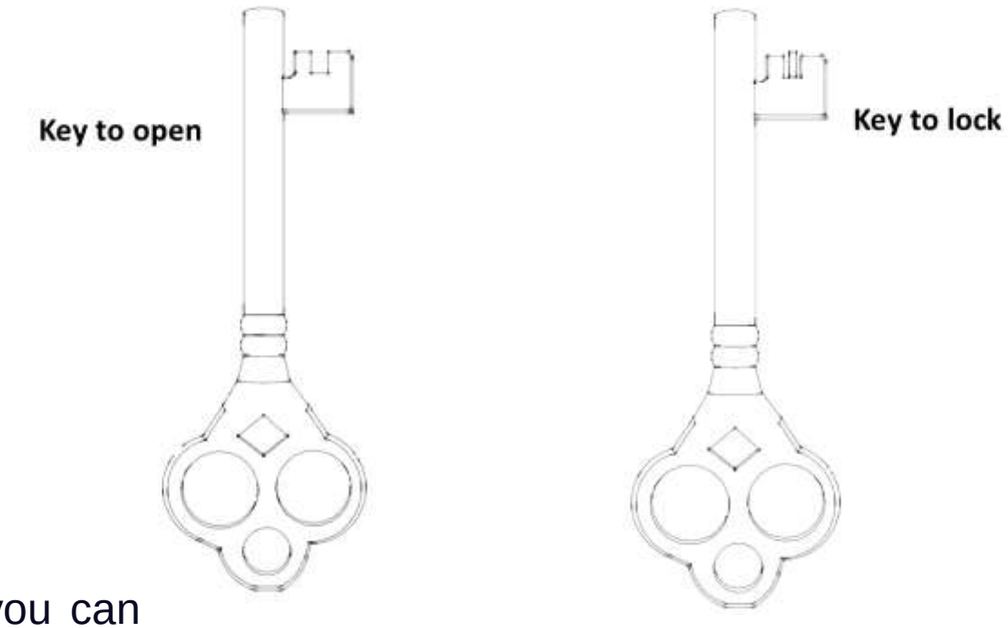
Asymmetric key encryption
Analogy

- Imagine you wanted to send something to your friend, but it was absolutely essential that nobody else, except your friend, could have access to that object. So, your friend buys an indestructible box, fabricated from the strongest metal on the planet, and sends it to you so that you can place the object in it. Your friend also sends you the key that can only be used to lock the box.

Now, this box has one more special property.
It has two keyholes. One keyhole to open the box, another to lock the box.



- Naturally, this box will also need two keys – one to open and another to lock it.

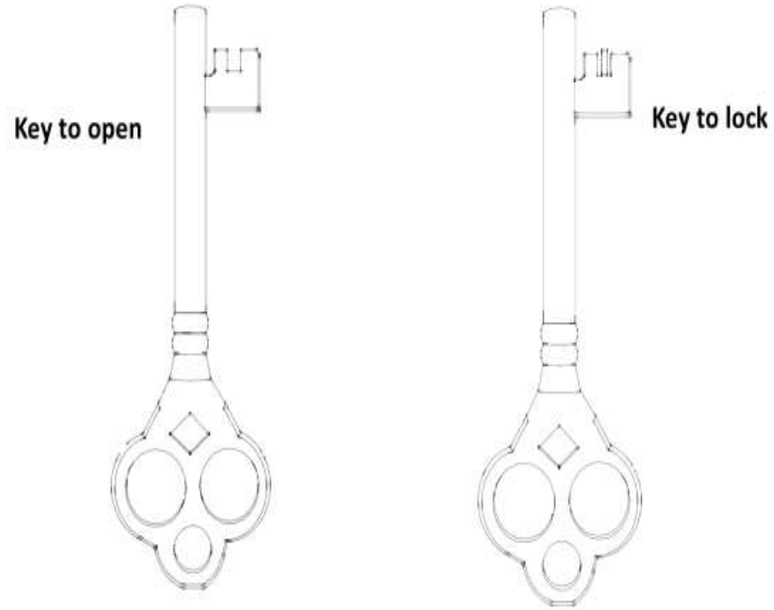


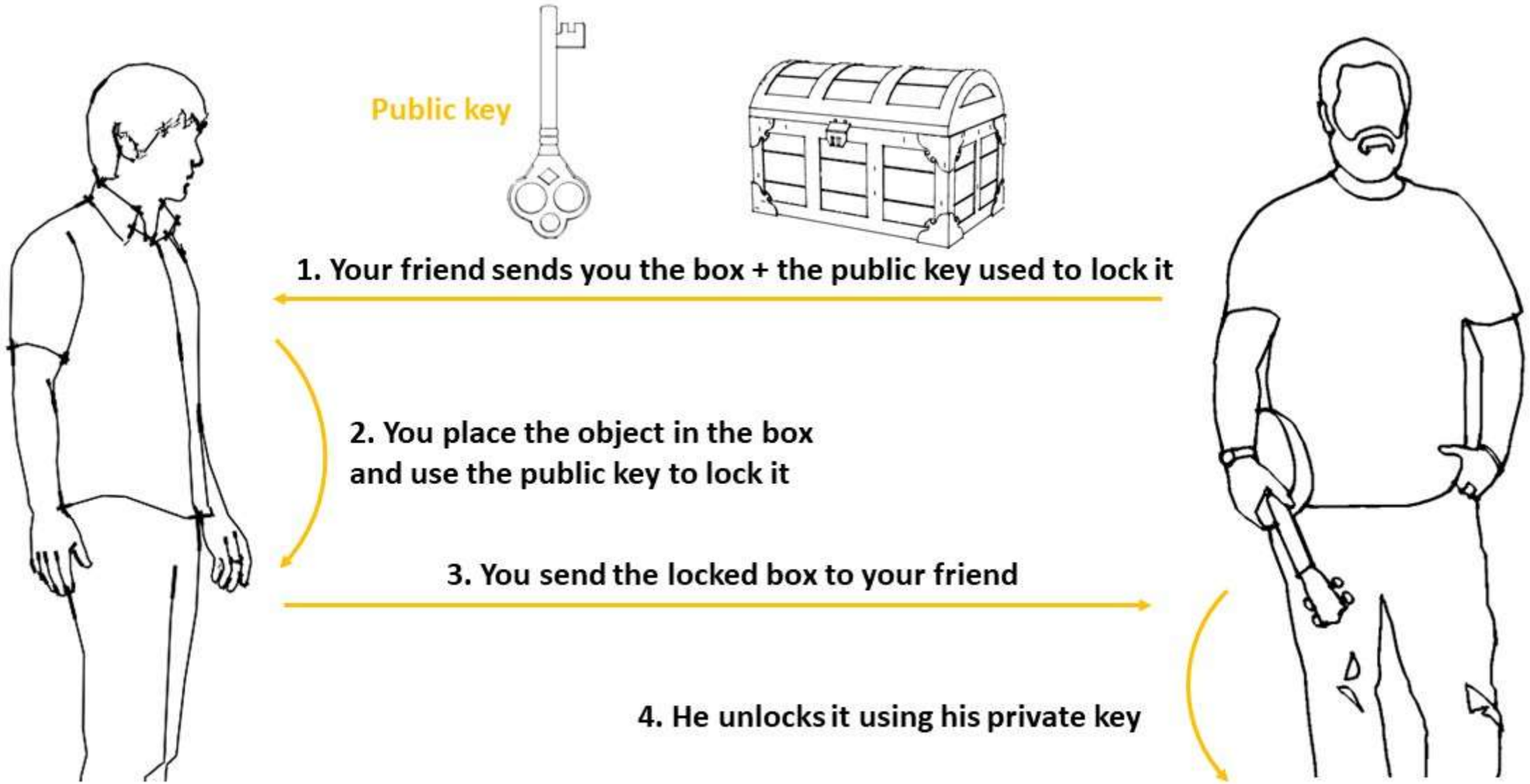
Both keys are similar, but not identical. As you can see in the image above, for example, the key used to open the box has two prongs while the key used to lock the box has three prongs.

- Naturally, this box will also need two keys – one to open and another to lock it.

As the sender of the object, all you have is the box to place the object in and a key to lock the box. Only your friend has the key that can unlock the box.

The key used to lock the box is called the public key, and cannot be used to open it, as that requires the private key. If anyone intercepted the package and made a copy of the public key, it could not be used to open the box, only to lock it. Only the person who holds the private key can open the box.




Public key
1. Your friend sends you the box + the public key used to lock it
2. You place the object in the box
and use the public key to lock it
3. You send the locked box to your friend
4. He unlocks it using his private key

Factors	Symmetric Key Cryptography	Asymmetric Key Cryptography
Size of cipher text	The same or smaller than the original plain text	The same or larger than the original plain text
Data size	Used for large amounts of data	Used for small amounts of data
Resource Utilization	Low	High
Key Lengths	128 or 256 bits	2048 or higher
Security	Less secure as only one key is used for both encryption and decryption	More secure as two keys are used, one for encryption and the other for decryption
Number of keys	One key for both encryption and decryption	Two keys, a public key and a private key, one for encryption and the other for decryption
Techniques	Provides confidentiality	Provides confidentiality, authenticity, and non-repudiation
Confidentiality	Only the key holder can decrypt the message	Only the private key holder can decrypt the message
Speed	Fast	Slow
Algorithms	Examples: 3DES, AES, DES and RC4	Examples: Diffie-Hellman, ECC, DSA, and RSA